

PERIPHERALS AND CONNECTIVITY

1. Connect five different types of peripherals (one each from input, output, storage, and communication categories) to your PC and document the connection method (USB, Bluetooth, Ethernet, etc.), including any visible device name in Device Manager or System Preferences.

Hint: Example peripherals: USB mouse, external hard drive, headphones, network adapter, webcam.

-> Peripheral Connection and Device Identification

The following table gives a suitable example of five peripherals connected to a desktop PC. Device names shown in Device Manager can vary depending on the manufacturer and Windows version.

No.	Peripheral	Category	Connection method	Example device name shown in Device Manager
1	USB Mouse	Input	USB-A	HID-compliant mouse
2	Webcam	Input	USB-A	USB Video Device
3	Headphones/Speakers	Output	3.5 mm audio jack	High Definition Audio Device
4	External HDD	Storage	USB 3.0/USB-A	USB Attached SCSI (UAS) Mass Storage Device
5	Network Adapter	Communication	Ethernet	Realtek PCIe GbE Family Controller

Documentation

1. USB Mouse — Input

Connected the mouse to a USB port. Windows detected it as a **HID-compliant mouse** in Device Manager.

2. Webcam — Input

Connected the webcam through USB. It appeared as a **USB Video Device** under Cameras/Imaging devices.

3. Headphones — Output

Connected the headphones to the 3.5 mm audio jack. The sound device appeared under **Sound, video and game controllers** as a High Definition Audio device.

4. External HDD — Storage

Connected the external hard drive through USB 3.0. Windows detected the storage controller/device as a **USB Attached SCSI (UAS) Mass Storage Device**, and the drive appeared in File Explorer.

5. Ethernet Network Adapter — Communication

Connected an Ethernet cable from the PC to the router/network switch. The adapter appeared as a **Realtek PCIe GbE Family Controller** under Network adapters.

2. Install drivers for a USB device manually by downloading the correct driver from the manufacturer's website, and then verify the device is working correctly in Device Manager (Windows) or System Information (Mac).

-> For a practical record, use the **exact model number of the USB device** before downloading its driver. The following is a suitable step-by-step procedure you can perform on Windows.

Manual USB Driver Installation

1. Identify the USB device

- Connect the USB device to the computer.
- Open **Device Manager** by pressing Windows + X → **Device Manager**.
- Locate the device, especially if it appears under **Other devices** with a yellow warning symbol.
- Note the **manufacturer and model number** of the device.

2. Download the correct driver

- Open the manufacturer's official website.
- Go to its **Support / Downloads / Drivers** section.
- Search for the exact device model and select the driver matching the installed Windows version and system architecture, such as Windows 11 64-bit.
- Download the driver package.

3. Install the driver manually

- Extract the downloaded ZIP file if necessary.
- Run the manufacturer's setup program, or open **Device Manager** → **right-click the USB device** → **Update driver** → **Browse my computer for drivers**.

- Select the folder containing the downloaded driver and complete the installation.
- Restart the computer if requested.

4. Verify the installation

- Open **Device Manager** again.
- Locate the USB device in its appropriate category.
- Confirm that the **yellow warning icon is no longer present**.
- Open **Properties → General** and check that the message says **“This device is working properly.”**
- Under **Driver**, record the driver provider, version, and date.

Example Record

Item	Observation
USB device	[Manufacturer + model]
Driver source	Official manufacturer website
Driver version	[Version shown in Device Manager]
Operating system	Windows [version] 64-bit
Installation method	Manual driver installation
Device Manager status	No warning symbol
Device status	This device is working properly

3. Set up a network printer (not USB) using its IP address on your local Wi-Fi or LAN, install the correct printer driver, and print a test page to confirm connectivity.

-> Network Printer Setup Using IP Address

Step-by-Step Procedure

1. Connect the printer to the network

- Turn on the printer.
- Connect it to the same **Wi-Fi network or LAN** as the computer.

- Obtain the printer's IP address from the printer's network settings or configuration/status page.

2. Open Windows printer settings

- Open **Settings → Bluetooth & devices → Printers & scanners**.
- Select **Add device**.

3. Add the printer by IP address

- Select **Add manually** when the printer is not automatically discovered.
- Choose **Add a printer using an IP address or hostname**.
- Select **TCP/IP Device** and enter the printer's IP address, for example: **192.168.1.100**
- Allow Windows to detect the network printer.

4. Install the correct printer driver

- Select the printer manufacturer and exact model.
- If Windows does not have the correct driver, download the driver package from the **manufacturer's official website**.
- Complete the driver installation.

5. Verify printer connectivity

- Go to **Printers & scanners** and select the installed printer.
- Open **Printer properties** and check the configured **IP address/port**.
- Use **Print Test Page** to verify communication between the PC and printer.

6. Confirm successful printing

- The test page should print correctly without an error.
- This confirms that the printer is reachable over the LAN/Wi-Fi and that the correct driver is installed.

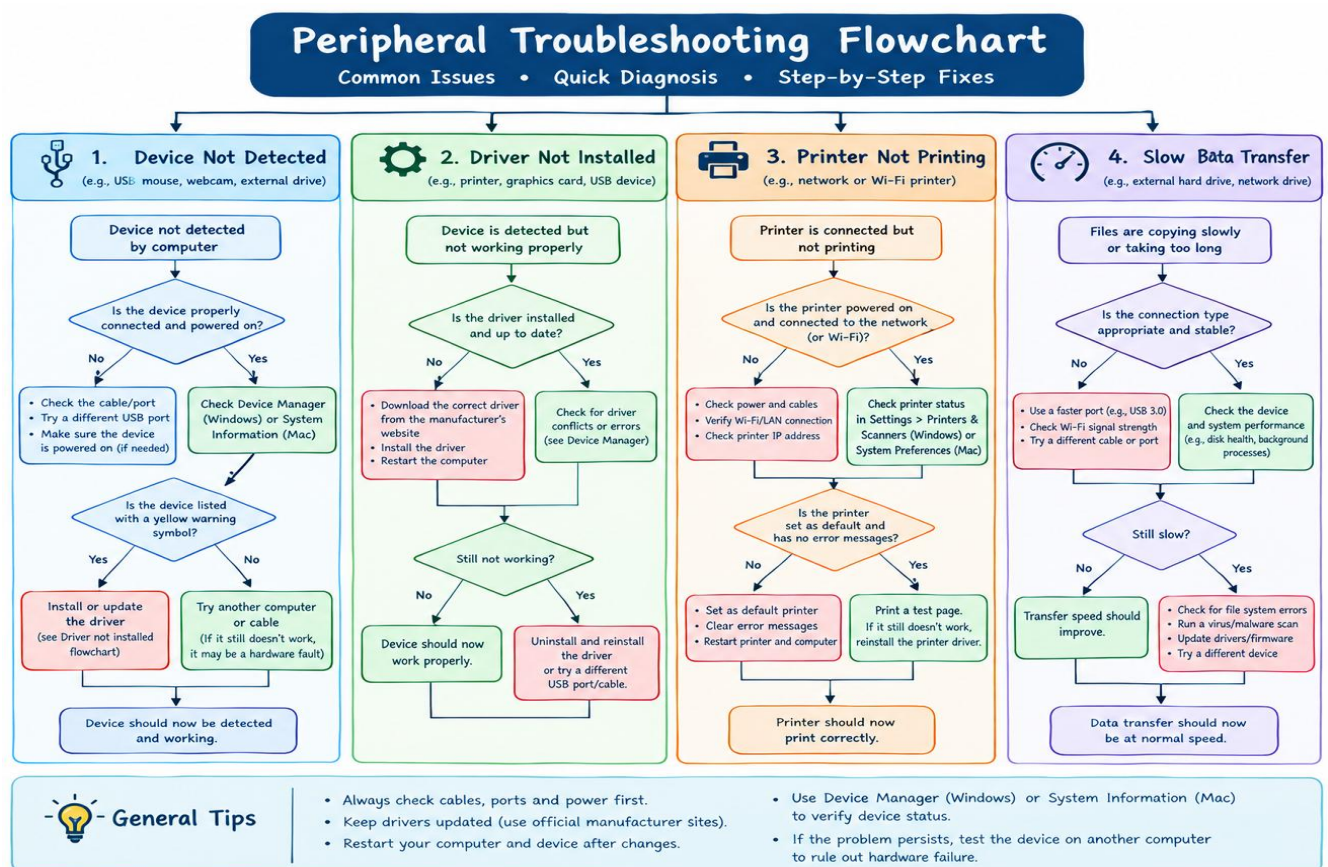
Example Lab Record

Item	Observation
Printer type	Network printer
Connection	Wi-Fi / Ethernet LAN

Item	Observation
Printer IP address	192.168.1.100
Port	Standard TCP/IP Port
Driver	Correct manufacturer/model driver
Driver installation	Successful
Test page	Printed successfully
Connectivity status	Connected / Working

4. Create a troubleshooting flowchart (hand-drawn or digital) that covers common peripheral issues such as 'Device not detected', 'Driver not installed', 'Printer not printing', and 'Slow data transfer', showing the steps to diagnose and solve each problem.

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5. Design a printer management plan for a startup office of 50 employees: specify how you would organize network printers, manage driver installations, and ensure everyone has access without printer queue conflicts.

Hint: Think about printer sharing, access permissions, and using printer queues or print servers.

-> **Printer Management Plan for a 50-Employee Startup Office**

For a 50-person office, I would use **network printers managed through a central print server** rather than connecting printers directly to individual PCs. This gives centralized control over drivers, permissions, and print queues.

1. Network Printer Organization

Printer	Location	Purpose	Access
Printer 1	Main office	General black-and-white printing	All employees
Printer 2	Accounts/HR	Confidential documents	Accounts + HR
Printer 3	Design/Operations	Color and high-quality printing	Design + Operations

Each printer would have a **fixed/reserved IP address** on the office LAN so that its address does not unexpectedly change.

2. Central Print Server

I would configure a **Windows print server** to manage all three printers.

Each printer would have its own queue, for example:

- Office-BW
- Accounts-Confidential
- Design-Color

Employees would connect to the shared printer queues rather than configuring individual IP connections. The server would manage jobs in the correct queue and prevent users from accidentally sending jobs to another department's printer.

3. Driver Management

The correct manufacturer drivers would be installed **once on the print server**.

When employees connect to a shared printer, the appropriate driver can be distributed to their computers through centralized management such as **Group Policy or device-management software**. This reduces inconsistent driver versions and makes updates easier.

4. Access Permissions

I would create groups such as:

- **All-Employees** → General office printer
- **Accounts-HR** → Confidential printer
- **Design-Operations** → Color printer

The print server would apply permissions so employees can only see/use printers appropriate for their role.

5. Preventing Queue Conflicts

Each physical printer would have **one centrally managed print queue**. Employees submit jobs to the queue instead of directly controlling the printer.

The print administrator could monitor stalled jobs, cancel duplicate jobs, and troubleshoot failed print jobs without affecting other printers.

6. Maintenance and Monitoring

The administrator should regularly check:

- Printer online/offline status
- Paper and toner levels
- Failed or stuck print jobs
- Driver and firmware updates
- Printer usage and departmental printing volume

Recommended Setup

50 employees → Central print server → Separate printer queues → Network printers

This setup provides centralized driver management, controlled access, and organized queues. It also makes adding or replacing a printer easier because employees continue using the same shared queue name rather than having to reconfigure every computer.